

SG-WAM: Self-Guided World Modeling in Geometry-Aware Policy Space

Anonymous submission

Abstract

World Action Models (WAMs) couple action generation with prediction of future states. Their effectiveness depends on whether future dynamics are modeled in a space that is both aligned with action generation and sufficiently geometry-aware to capture where and how actions change the scene. Existing WAMs typically satisfy only part of this requirement, relying on either perceptually heavy observation-space targets or auxiliary latent spaces that are not jointly structured for action relevance and geometry. We propose **SG-WAM**, a self-guided framework that learns geometry-aware action-conditioned dynamics directly in the policy-derived representation space. SG-WAM introduces learnable dynamics tokens and a *Self-Guided World Predictor* that forecasts their future latent states conditioned on intervening robot actions. Prediction targets are generated by an exponential moving average copy of the same policy backbone, providing stable supervision within the representation family used by the action expert. Geometric supervision further structures the policy image-token representations, providing spatially grounded context for the dynamics tokens and yielding a future-alignment space that is both action-relevant and geometry-aware. Latent future prediction, geometric grounding, and flow-matching action generation are jointly optimized end-to-end in a unified framework. Built on a 0.9B model without large-scale embodied pretraining, SG-WAM achieves 98.5% average success on LIBERO and 73% on LIBERO-Plus, while outperforming strong baselines in both in-distribution and out-of-distribution real-world evaluations.

1 Introduction

World Action Models (WAMs) (Guo et al. 2026; Yuan et al. 2026; Cen et al. 2025b; Su et al. 2026; Sun et al. 2026; Bu et al. 2025; Zhang et al. 2026a; Chen et al. 2026c) have recently emerged as a promising paradigm for robot manipulation by coupling action generation with prediction of future states. Unlike conventional policies that directly map observations to actions, WAMs explicitly supervise the policy to anticipate how robot interactions alter the environment and advance task progress. Such future-oriented supervision can encourage policy representations to capture action-induced dynamics that are weakly constrained by action imitation

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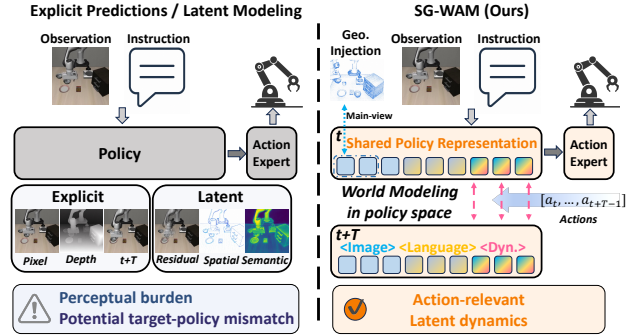


Figure 1: Conceptual comparison of existing WAMs and SG-WAM. Whereas explicit and auxiliary latent targets may introduce perceptual burden or target-policy mismatch, SG-WAM learns intervening-action-conditioned dynamics in geometry-structured, policy-derived representations.

alone. However, the effectiveness of WAMs critically depends on the representation space in which future supervision is imposed and how closely this space is coupled to action generation.

Existing WAMs (Cen et al. 2025b,a; Zhang et al. 2026a; Sun et al. 2026; Yuan et al. 2026; Hu et al. 2024) broadly model action-conditioned futures in either observation space or latent representation space. Observation-space approaches predict future images, videos, depth maps, or other perceptual signals to provide dense supervision of scene evolution (Cen et al. 2025b,a; Yang et al. 2026). Although informative, these targets require the model to capture variations in texture, viewpoint, background, etc., that may be only weakly related to the state changes governing successful manipulation. Consequently, substantial modeling capacity may be allocated to perceptual fidelity rather than manipulation-relevant dynamics.

Latent WAMs instead model future information in representation space, reducing the burden of reconstructing manipulation-irrelevant appearance details. Existing methods instantiate this idea in different ways: some directly predict future visual features or align policy representations with features extracted from future observations (Lyu et al. 2026; Zheng et al. 2025; Zhang et al. 2026a), while others construct compact predictive representations designed to capture action-relevant feature changes or subspaces (Sun et al. 2026;

Su et al. 2026). Despite their differences, these methods commonly impose future-oriented supervision through externally defined or auxiliary target representations that are not explicitly guaranteed to coincide with the representations directly used to condition action generation. This creates a potential mismatch between the information emphasized by the predictive objective and that required by the action expert.

Such limitations suggest that simply moving future prediction from observation space to latent space is insufficient. Although action-oriented representations can suppress appearance variation, such compression does not guarantee preservation of the fine-grained spatial information required for manipulation. An ideal future-modeling space must therefore not only be directly coupled to action generation, but also retain the geometry needed to represent where and how actions change the scene. This is critical because successful manipulation depends not only on recognizing what changes but also on capturing the spatial details. However, existing WAMs typically encode geometry through perceptual reconstruction, auxiliary targets, or predefined feature spaces (Zhen et al. 2025; Zhang et al. 2026a; Guo et al. 2026; Sun et al. 2026; Tian et al. 2026), rather than directly using it to structure the policy-coupled future-alignment space. This raises a central challenge: *how to construct a geometry-aware future-alignment space that is directly coupled to the representations used for action generation.*

Contribution. To address this challenge, we propose **SG-WAM**, a self-guided framework for learning geometry-aware action-conditioned dynamics directly in the policy representation space used by the action expert, as illustrated in Figure 1. Rather than reconstructing future observations or predicting features from a separate future encoder, SG-WAM inserts learnable dynamics tokens into the VLM sequence and employs a *Self-Guided World Predictor* to forecast their future states conditioned on intervening robot actions. The prediction targets are generated by an exponential moving average (EMA) copy of the same policy backbone, providing stable supervision within the same representation family used for action generation. To ensure that this policy-coupled future-alignment space retains the spatial structure required for manipulation, SG-WAM further uses geometric supervision to explicitly shape the policy image-token representations. These geometry-grounded image tokens provide spatial context for the dynamics tokens, yielding a latent dynamics space that is jointly action-relevant and geometry-aware. Latent future prediction, geometric grounding, and flow-matching action generation are optimized end-to-end in a unified training framework, allowing future supervision to directly organize the representations consumed by the action expert, while requiring only the current observation and language instruction at inference.

Results. Built on a 0.9B model without large-scale embodied pretraining, SG-WAM achieves 98.5% average success on LIBERO and 73% on LIBERO-Plus, with clear gains over strong baselines in both in-distribution and out-of-distribution real-world evaluations.

2 Related Works

WAMs with Explicit Future Modeling. Recent World Action Models extend Vision-Language-Action policies (Bjorck et al. 2025; Black et al. 2024, 2025; Kim et al. 2024; Zhang et al. 2026b) by predicting how the environment evolves under robot actions (Ye et al. 2026; Li et al. 2026; Hu et al. 2024; Cen et al. 2025b). A representative line of work generates future images or videos as intermediate targets for action generation. UniPi (Du et al. 2023) predicts task-conditioned videos and recovers actions through inverse dynamics, while subsequent methods improve long-horizon reasoning and controllability through vision-language planning, hierarchical decomposition, and stronger video generation models (Du et al. 2024; Long et al. 2026; Yang et al. 2025; Wan et al. 2025; Seedance et al. 2026). Recent approaches further enrich predicted futures with depth, surface normals, or 4D structure (Zhen et al. 2025; Guo et al. 2026; Zhou et al. 2026; Wang et al. 2026), while structured representations such as optical flow and motion trajectories provide alternative interfaces between prediction and control (Ko et al. 2024; Xu et al. 2024; Zhi et al. 2025; Bharadhwaj et al. 2024). Although explicit prediction provides dense and interpretable supervision, observation-space objectives also require modeling texture, background, illumination, and viewpoint variations that may be weakly related to manipulation.

WAMs with Implicit Latent Modeling. To avoid reconstructing detailed observations, another line of work models future dynamics in compact latent spaces. These methods learn predictive representations of scene evolution and integrate them with policy learning (Chen et al. 2025; Bu et al. 2025; Ye et al. 2025; Chen et al. 2026b; Yuan et al. 2026). Joint-embedding predictive models further show that future information can be learned without pixel-level reconstruction (Assran et al. 2025; Maes et al. 2026). In robot learning, latent future modeling has been instantiated through latent actions, predicted future features, imagined trajectories, and compact condition spaces (Bruce et al. 2024; Gao et al. 2026; Zhu et al. 2025; Zheng et al. 2025; Su et al. 2026; Chen et al. 2026c; Sun et al. 2026). Recent methods further explore future-informed latent queries, latent visual subgoals in pre-trained feature spaces, and jointly modeled spatial-temporal latents and actions (Luo et al. 2026; Chen et al. 2026a; Tian et al. 2026). Unified architectures increasingly couple predictive states with action generation through shared or closely connected policy pathways (Bi et al. 2025; Lyu et al. 2026).

However, jointly training prediction and control does not necessarily mean that the future target is constructed from the same policy used for action generation. Existing WAMs obtain future supervision from future-observation or target encoders (Zheng et al. 2025; Sun et al. 2026), pretrained feature spaces or latent visual subgoals (Chen et al. 2026a), future-informed posterior branches (Luo et al. 2026), compact condition spaces (Su et al. 2026), or jointly modeled spatial-temporal latents and actions (Tian et al. 2026). In contrast, SG-WAM constructs its current predictive state and future target from online and EMA copies of the same policy. Geometric supervision further shapes the policy visual context from which the dynamics-token states are formed, coupling spatial

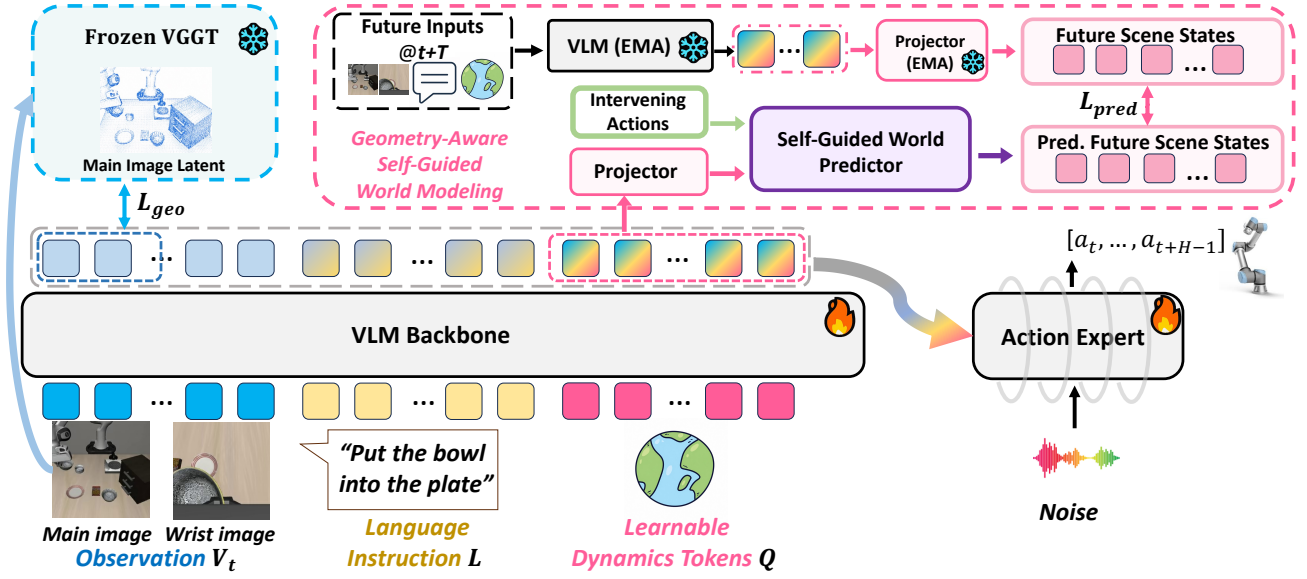


Figure 2: Overview of the SG-WAM framework. The VLM jointly contextualizes multi-view observations, language, and learnable dynamics tokens for action generation. During training, a frozen VGGT teacher shapes main-view image tokens, while SGWP predicts future dynamics-token states conditioned on intervening actions and aligns them with an EMA policy target. All auxiliary branches are removed at inference.

grounding, action-conditioned latent prediction, and action generation through shared policy representations.

3 Methodology

3.1 Problem Formulation

We seek to build a World Action Model (WAM) that jointly learns to generate robot actions and model how the environment evolves under those actions. Given the current multi-view observation $\mathbf{V}_t$ and task instruction $\mathbf{L}$, the model predicts an action chunk $\hat{\mathbf{A}}_t$ while learning how a latent policy state evolves under an intervening action sequence. During training, the expert action sequence

$$\mathbf{A}_t^{(\Delta)} = [\mathbf{a}_t, \dots, \mathbf{a}_{t+\Delta-1}] \quad (1)$$

conditions this transition, while the future observation $\mathbf{V}_{t+\Delta}$ provides its target.

Let $\mathbf{z}_t^Q$ denote the latent dynamics representation associated with the current policy state, and let $\mathbf{z}_{t+\Delta}^Q$ denote its future target representation. We model their action-conditioned transition as

$$\hat{\mathbf{z}}_{t+\Delta}^Q = \mathcal{F}_\psi(\mathbf{z}_t^Q, \mathbf{e}_t^A), \quad \hat{\mathbf{z}}_{t+\Delta}^Q \approx \text{sg}(\mathbf{z}_{t+\Delta}^Q), \quad (2)$$

where $\mathbf{e}_t^A$ denotes the feature representation of $\mathbf{A}_t^{(\Delta)}$, $\mathcal{F}_\psi$ denotes the action-conditioned transition model and $\text{sg}(\cdot)$ denotes the stop-gradient operation.

Unlike formulations that define the future target in observation space or through a separate future encoder, we construct both $\mathbf{z}_t^Q$ and $\mathbf{z}_{t+\Delta}^Q$ from the same policy representation family. The following sections describe how these dynamics representations are constructed from policy token states, geometrically shaped, and used for both future prediction and action generation.

3.2 Overall Framework

SG-WAM realizes this formulation through two complementary designs. First, self-guided prediction models the action-conditioned evolution of policy-derived dynamics tokens using targets generated by the same policy pathway. Second, geometry supervision shapes the visual-token context from which these dynamics states are formed, encouraging the shared policy representation to preserve the spatial information required for manipulation.

As illustrated in Figure 2, SG-WAM performs action-conditioned world modeling within the internal policy representation space. The visual, language, and learnable dynamics tokens are jointly contextualized within a shared VLM backbone (Qwen3.5-0.8B) (Team 2026). Geometry supervision acts on the policy visual tokens, while the Self-Guided World Predictor models the action-conditioned evolution of the resulting dynamics-token states. These dynamics-token states remain in the context consumed by the action expert, allowing future prediction and action generation to jointly optimize the same policy representation.

Given the current multi-view observation $\mathbf{V}_t$ and language instruction $\mathbf{L}$, they are first encoded into the corresponding representations $\mathbf{f}_t^V \in \mathbb{R}^{N_v \times D}$ and $\mathbf{f}_t^L \in \mathbb{R}^{N_l \times D}$, where N_v and N_l are the number of visual and language tokens, and D is the feature dimension. We append N_q learnable dynamics tokens $\mathbf{Q} \in \mathbb{R}^{N_q \times D}$ to the visual and language tokens and process them jointly through the VLM backbone $\mathcal{E}_\theta$ to get the last-layer contextualized token representations $\mathbf{H}_t$, which include visual, language and dynamics token states:

$$\mathbf{H}_t = \mathcal{E}_\theta([\mathbf{f}_t^V, \mathbf{f}_t^L, \mathbf{Q}]), \quad \mathbf{H}_t = [\mathbf{H}_t^V, \mathbf{H}_t^L, \mathbf{H}_t^Q]. \quad (3)$$

Here, $\mathbf{H}_t^V$, $\mathbf{H}_t^L$, and $\mathbf{H}_t^Q$ denote the contextualized visual,

language, and dynamics-token states, respectively.

To construct a predictive space suitable for robust manipulation, a frozen 3D foundation model (Wang et al. 2025) geometrically shapes the VLM main-view visual-token states $\mathbf{H}_t^{V,m} \in \mathbb{R}^{N_v^m \times D}$ during training, where N_v^m denotes the number of main-view visual tokens. The Self-Guided World Predictor (SGWP) then takes the dynamics-token states $\mathbf{H}_t^Q \in \mathbb{R}^{N_q \times D}$, which are contextualized by the geometry-shaped visual states, together with the intervening action sequence $\mathbf{A}_t^{(\Delta)}$ to predict the future latent states. These predictions are aligned with the target EMA states (Grill et al. 2020) produced by feeding the future observation to an EMA copy of the policy.

Moreover, the contextualized VLM states $\mathbf{H}_t$ are also provided to a flow-matching action expert to generate the continuous action chunk $\hat{\mathbf{A}}_t$. During inference, the frozen geometry teacher, the SGWP prediction branch, and the EMA target pathway are removed. The online VLM, learnable dynamics tokens, and action expert form the deployed policy.

3.3 Self-Guided World Modeling

Geometry-Aware Policy States. Although policy-derived latent representations are directly coupled to action generation, this coupling alone does not ensure that they preserve the local spatial information required for manipulation. We therefore use a frozen geometry model as an auxiliary teacher for the policy visual tokens. Importantly, the teacher features do not define the future-prediction target; they only shape the visual context from which the dynamics-token states are formed.

Given the current main-view observation $\mathbf{V}_t^m$, a frozen VGGT model extracts its final-layer geometric features, which are spatially pooled to obtain $\mathbf{Z}_t^G \in \mathbb{R}^{N_v^m \times D_g}$, where D_g denotes the VGGT feature dimension. On the policy side, a projector $\mathcal{G}_\gamma$ first maps the VLM last hidden states corresponding to the main-view visual tokens $\mathbf{H}_t^{V,m}$ to the teacher feature dimension $\hat{\mathbf{Z}}_t^G \in \mathbb{R}^{N_v^m \times D_g}$. We then align $\hat{\mathbf{Z}}_t^G$ with $\mathbf{Z}_t^G$ using the cosine similarity objective:

$$\mathcal{L}_{\text{geo}} = \frac{1}{N_v^m} \sum_{j=1}^{N_v^m} \left[1 - \cos \left(\hat{\mathbf{Z}}_{t,j}^G, \mathbf{Z}_{t,j}^G \right) \right]. \quad (4)$$

Because the dynamics tokens are jointly contextualized with these geometry-grounded visual states, $\mathcal{L}_{\text{geo}}$ provides a pathway for spatial information to enter the policy-derived dynamics representation used for future prediction. See Appendix for more details.

Self-Guided World Predictor. Built on the geometry-grounded policy context, the dynamics-token states serve as compact carriers for modeling action-conditioned scene evolution. We introduce the Self-Guided World Predictor (SGWP) to model how the scene state evolves under robot actions. The key distinction of the SGWP lies in its target construction. Instead of predicting future features defined by a separate encoder, both the current dynamics representation and its future target are derived from the same policy architecture. The online branch produces the current dynamics-token states, while an EMA copy of the policy produces a slowly

evolving target from the future observation, allowing latent prediction to directly shape the policy representations used for action generation.

For the online pathway, the current dynamics-token states $\mathbf{H}_t^Q \in \mathbb{R}^{N_q \times D}$ are mapped into the compact latent prediction space by the projector $\mathbf{z}_t^Q = \mathcal{P}_\phi(\mathbf{H}_t^Q)$, $\mathbf{z}_t^Q \in \mathbb{R}^{N_q \times d}$, where ϕ denotes the parameters of the projector and d is the space dimension. The intervening action sequence $\mathbf{A}_t^{(\Delta)}$ is encoded by an action encoder $\mathcal{A}_\eta$ as $\mathbf{e}_t^A = \mathcal{A}_\eta(\mathbf{A}_t^{(\Delta)})$. Conditioned on the current dynamics representation and the intervening actions, SGWP predicts the future dynamics representation:

$$\hat{\mathbf{z}}_{t+\Delta}^Q = \mathcal{F}_\psi \left(\mathbf{z}_t^Q, \mathbf{e}_t^A \right), \quad \hat{\mathbf{z}}_{t+\Delta}^Q \in \mathbb{R}^{N_q \times d}, \quad (5)$$

where $\mathcal{F}_\psi$ denotes the action-conditioned predictor. See Appendix for more details of SGWP.

The prediction target is generated from the future observation $\mathbf{V}_{t+\Delta}$ and the same language instruction $\mathbf{L}$ by an EMA copy of the same policy pathway. Specifically, the target VLM processes the future visual tokens $\mathbf{f}_{t+\Delta}^V \in \mathbb{R}^{N_v \times D}$, language tokens $\mathbf{f}^L \in \mathbb{R}^{N_l \times D}$, and EMA dynamics tokens $\bar{\mathbf{Q}} \in \mathbb{R}^{N_q \times D}$ to obtain

$$\bar{\mathbf{H}}_{t+\Delta}^Q = [\mathcal{E}_{\bar{\theta}}([\mathbf{f}_{t+\Delta}^V, \mathbf{f}^L, \bar{\mathbf{Q}}])]^Q, \quad (6)$$

where $[\cdot]^Q$ selects the states corresponding to the dynamics tokens. The target states are then projected into the same latent prediction space: $\bar{\mathbf{z}}_{t+\Delta}^Q = \mathcal{P}_{\bar{\phi}}(\bar{\mathbf{H}}_{t+\Delta}^Q)$, $\bar{\mathbf{z}}_{t+\Delta}^Q \in \mathbb{R}^{N_q \times d}$. The target parameters are updated by

$$\bar{\Theta} \leftarrow \mu \bar{\Theta} + (1 - \mu) \Theta, \quad (7)$$

where $\Theta = (\theta, \phi, Q)$ and $\bar{\Theta} = (\bar{\theta}, \bar{\phi}, \bar{Q})$ denote the online and EMA parameters of the VLM $\mathcal{E}$, the projector $\mathcal{P}$, and the dynamics tokens, and μ is the EMA momentum coefficient. The target pathway is detached from gradient backpropagation and updated only through the EMA rule.

The latent prediction objective minimizes the mean squared error (MSE) between the predicted and EMA future states:

$$\mathcal{L}_{\text{pred}} = \frac{1}{N_q d} \left\| \hat{\mathbf{z}}_{t+\Delta}^Q - \text{sg} \left(\bar{\mathbf{z}}_{t+\Delta}^Q \right) \right\|_F^2, \quad (8)$$

where $\text{sg}(\cdot)$ denotes stop-gradient. Gradients from $\mathcal{L}_{\text{pred}}$ update the online VLM, dynamics tokens, projector, action encoder, and SGWP, while leaving the EMA target pathway gradient-free. Because the same dynamics-token states remain in the context supplied to the action expert, the prediction and action objectives jointly optimize the shared policy backbone.

3.4 Conditional Flow-Matching Action Expert

The action expert is conditioned on the complete policy context $\mathbf{H}_t = [\mathbf{H}_t^V, \mathbf{H}_t^L, \mathbf{H}_t^Q]$, such that the dynamics-token states optimized by future prediction remain part of the context used for action generation. We adopt conditional flow matching to generate an H -step action chunk. Given an expert action chunk $\mathbf{A}_t$, Gaussian noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and $\tau \sim \mathcal{U}(0, 1)$, we define $\mathbf{A}_t^\tau = \tau \mathbf{A}_t + (1 - \tau) \epsilon$ and optimize

$$\mathcal{L}_{\text{act}} = \mathbb{E}_{\mathbf{A}_t, \epsilon, \tau} \left[\left\| v_\omega(\mathbf{A}_t^\tau, \tau, \mathbf{H}_t) - (\mathbf{A}_t - \epsilon) \right\|_2^2 \right]. \quad (9)$$

At inference, the action chunk is obtained by integrating the learned velocity field while conditioning on $\mathbf{H}_t$.

Method	Params	Embodied PT.	Spatial	Object	Goal	Long	Avg.
OpenVLA-OFT (Kim, Finn, and Liang 2025)	7B	✓	97.6	98.4	97.9	94.5	97.1
π_0 (Black et al. 2024)	3.3B	✓	98.0	96.8	94.4	88.4	94.4
π_0 -FAST (Pertsch et al. 2025)	3.3B	✓	96.4	96.8	88.6	60.2	85.5
$\pi_{0.5}$ (Black et al. 2025)	3.3B	✓	98.8	98.2	98.0	92.4	96.9
GR00T N1.6 (Bjorck et al. 2025)	3B	✓	97.7	98.5	97.5	94.4	97.0
Spatial Forcing (Li et al. 2025)	7B	✓	99.4	99.6	98.8	<u>96.0</u>	98.5
WorldVLA (Cen et al. 2025b)	7B	✗	87.6	96.2	83.4	60.0	81.8
LAPA (Ye et al. 2025)	7B	✓	55.4	58.8	74.6	73.8	65.7
RynnVLA-002 (Cen et al. 2025a)	7B	✗	<u>99.0</u>	<u>99.8</u>	96.4	94.4	97.4
Mantis (Yang et al. 2026)	5.8B	✓	98.8	99.2	94.4	94.2	96.7
UniVLA (Bu et al. 2025)	7B	✓	96.5	96.8	95.6	92.0	95.2
Fast-WAM (Yuan et al. 2026)	6B	✗	98.2	100.0	97.0	95.2	<u>97.6</u>
VLA-JEPA (Sun et al. 2026)	2B	✗	94.8	99.6	95.8	94.0	96.1
SG-WAM	0.9B	✗	99.4	<u>99.8</u>	<u>98.6</u>	96.2	98.5

Table 1: Simulation results on LIBERO. Success rates are reported for the four standard LIBERO suites. **Bold** indicates the best result, and underlining indicates the second-best result.

Method	Params	Camera	Robot	Language	Light	Background	Noise	Layout	Overall
WorldVLA (Cen et al. 2025b)	7B	0.1	27.9	41.6	43.7	17.1	10.9	38.0	25.0
Spatial Forcing (Li et al. 2025)	7B	20.1	13.4	40.9	29.1	33.4	25.7	39.3	29.1
Mantis (Yang et al. 2026)	5.8B	15.7	41.8	45.9	45.1	28.9	39.2	62.5	39.8
UniVLA (Bu et al. 2025)	7B	4.3	<u>50.3</u>	71.8	59.1	80.0	25.3	34.3	41.5
Fast-WAM (Yuan et al. 2026)	6B	16.4	44.5	68.9	78.2	53.7	37.7	60.7	50.0
π_0 (Black et al. 2024)	3.3B	13.8	6.0	58.8	85.0	81.4	<u>79.0</u>	68.9	53.6
VLA-JEPA (Sun et al. 2026)	2B	40.3	55.7	72.9	88.2	70.5	38.2	74.6	62.9
OpenVLA-OFT (Kim, Finn, and Liang 2025)	7B	<u>56.4</u>	31.9	<u>79.5</u>	<u>88.7</u>	93.3	75.8	<u>74.2</u>	<u>69.6</u>
SG-WAM	0.9B	58.6	48.9	81.4	89.8	<u>86.1</u>	80.7	<u>74.2</u>	73.0

Table 2: Zero-shot transfer results on LIBERO-Plus. Success rates are reported under different perturbation settings. **Bold** indicates the best result, and underlining indicates the second-best result.

3.5 Training Objective

The model is jointly trained with

$$\mathcal{L} = \mathcal{L}_{\text{act}} + \lambda_{\text{geo}}\mathcal{L}_{\text{geo}} + \lambda_{\text{pred}}\mathcal{L}_{\text{pred}}, \quad (10)$$

where λ_{geo} and λ_{pred} weight the geometry-shaping and action-conditioned latent prediction losses. The action loss updates the action expert and the shared online policy backbone, including the dynamics tokens. The prediction loss additionally updates the prediction projector, action encoder, and SGWP, while the geometry loss updates the geometry projector and shapes the online visual-token states. The VGGT teacher remains frozen, and the target policy pathway is updated only through EMA. During inference, the geometry teacher, SGWP prediction branch, and EMA target pathway are removed. All trainable online components are jointly optimized in a single training stage.

4 Experiments

To evaluate the generalization and robustness of SG-WAM, we conduct comprehensive simulation and real-world experiments. We use two simulation benchmarks, LIBERO and LIBERO-Plus, and further evaluate the policy on real-world robot manipulation tasks. We compare SG-WAM with representative baselines to assess its performance across both

simulated and physical environments. We provide more details of simulation and real-world experiments in the appendix.

4.1 Simulation Setup and Baselines

LIBERO is a standard benchmark for robot manipulation. It contains four task suites, LIBERO-Spatial, LIBERO-Object, LIBERO-Goal, and LIBERO-Long, which evaluate spatial reasoning, object-centric manipulation, goal-conditioned behavior, and long-horizon task execution. LIBERO-Plus extends this setting with more challenging task configurations and visual variations, providing a stronger test of policy robustness and generalization. For LIBERO, we jointly train the policy on all four standard suites and evaluate it on the corresponding benchmark tasks. For LIBERO-Plus, we use the policy trained on LIBERO and directly evaluate it without fine-tuning, forming a zero-shot transfer setting. This protocol allows us to assess both in-domain manipulation performance and out-of-distribution generalization under more challenging task variations.

We compare SG-WAM with representative baselines, including generalist pretrained policies (Kim, Finn, and Liang 2025; Black et al. 2024, 2025; Pertsch et al. 2025; Bjorck et al. 2025; Li et al. 2025), explicit WAMs (Cen et al. 2025b,a; Yang et al. 2026) and implicit WAMs (Ye et al. 2025; Bu et al.

2025; Yuan et al. 2026; Sun et al. 2026). These comparisons cover the main families of recent approaches. Simulation baseline results are reported from the corresponding papers under their stated LIBERO and LIBERO-Plus protocols.

4.2 Simulation Results

Table 1 reports the results on the four standard LIBERO suites. SG-WAM achieved an average success rate of 98.5%, matching the strongest baseline while using a substantially smaller 0.9B model and no additional embodied pretraining. In contrast, most high-performing baselines rely on larger backbones or large-scale embodied pretraining. LIBERO-Plus evaluates zero-shot robustness under distribution shifts in camera viewpoint, robot embodiment, language, illumination, background, observation noise, and scene layout. As shown in Table 2, SG-WAM achieves the highest overall success rate of 73.0%, demonstrating the strongest overall zero-shot transfer performance across these perturbations. Although Spatial Forcing and Fast-WAM are among the strongest methods on standard LIBERO, their performance decreases substantially under the distribution shifts introduced by LIBERO-Plus. In contrast, SG-WAM achieves the best performance under camera, language, illumination, and layout shifts while remaining competitive under background and observation-noise perturbations, suggesting that its learned policy representations are less sensitive to changes in visual and linguistic input distributions. These results indicate that SG-WAM provides a stronger balance between in-distribution task performance and out-of-distribution robustness.

4.3 Real-World Experimental Setup and Baselines

We construct the in-distribution (ID) setting using three tasks and collect 100 expert demonstrations for each task: (1) **Pick and Place**. The robot retrieves a blue cube from an open drawer and places it into a bowl while a barrier obstructs the direct transfer path. (2) **Towel Folding**. The robot folds a flat towel in half twice to produce a compact configuration. (3) **Toolbox Organization**. The robot sequentially places a screwdriver and two gears into a toolbox before closing it. Figure 3 shows the visualization of the tasks. To evaluate generalization, we assess out-of-distribution (OOD) generalization under three conditions not presented in the expert data: **Background Shift**, **Light Change**, and **Novel Object**.

We use a UR5e robotic arm as our main manipulation platform, an Azure Kinect camera to capture main RGB image, and a RealSense D405 to capture robot gripper RGB image. The real experimental platform is shown in Figure 3. We select VPP (Hu et al. 2024) and VLA-JEPA (Sun et al. 2026) as representative baselines for our real-world experiments, covering explicit and implicit world-modeling paradigms, respectively. VPP explicitly models future visual observations through video prediction, whereas VLA-JEPA predicts future representations in latent space without reconstructing observations.

For the real-world evaluation, we test each method for 20 trials on Pick and Place and Towel Folding tasks. Because the Toolbox Organization task requires substantially longer execution, we conduct 10 trials per method on this task. For

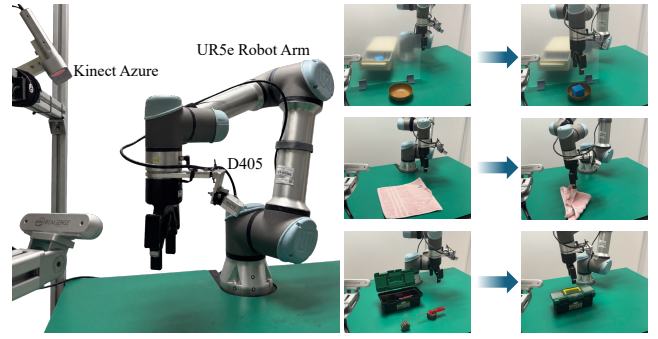


Figure 3: **Left:** Visualization of the real-world platform. We use a UR5e robot arm as the manipulation platform, the Kinect Azure camera as the main camera and the RealSense D405 as the gripper camera. **Right:** Visualization of 3 real-world tasks. 1) **Top:** Pick and Place. 2) **Middle:** Towel Folding. 3) **Bottom:** Toolbox Organization.

each trial, the initial scene is randomized and reproduced as closely as possible across methods to ensure a fair comparison.

4.4 Real-World Results

As shown in Table 3, SG-WAM consistently outperformed VPP and VLA-JEPA under both in-distribution (ID) and out-of-distribution (OOD) conditions. It achieved the highest ID success rate across all three tasks and retained its advantage under changes in background, light, and object. This advantage also extended to the long-horizon Toolbox Organization task, whose multi-stage structure increases the risk of compounding execution errors. These results indicate that SG-WAM supports reliable execution under familiar conditions while remaining robust to visual distribution shifts and extended manipulation horizons.

4.5 Ablation Study

We ablate geometric supervision and self-guided world modeling on the four LIBERO suites. As shown in Table 4, the model without either component achieves an average success rate of 95.3%. Combining both components yields the best average performance of 98.5%. Removing world modeling reduces the average success rate by 1.9 percentage points, compared with a 0.9-percentage-point reduction after removing geometric supervision. This difference is most pronounced on LIBERO-Long, where removing world modeling decreases performance from 96.2% to 92.2%. These results provide empirical support for the effectiveness of self-guided world modeling in action generation, especially on tasks that require coherent modeling of long-horizon state transitions. Geometric supervision grounds the learned representations in manipulation-relevant spatial structure, thereby facilitating the organization of action-conditioned transitions. Together, the two objectives produce the strongest performance, particularly on LIBERO-Long, suggesting that spatial grounding and predictive supervision provide complementary benefits.

We further varied the number of learnable dynamics tokens while keeping all other settings fixed. Increasing the number of tokens from one to eight improved the average success rate

Model	Pick and Place				Towel Folding				Toolbox Organization
	ID	Background	Light Change	Novel Object	ID	Background	Light Change	Novel Object	ID
VLA-JEPA	35%	20%	25%	20%	20%	10%	10%	15%	20%
VPP	30%	15%	10%	10%	35%	15%	15%	10%	30%
SG-WAM	75%	55%	60%	40%	45%	25%	35%	25%	50%

Table 3: Success rates under different visual perturbations. **Bold** indicates the best result.

Geo.	WM.	Spatial	Object	Goal	Long	Avg.
✗	✗	97.0	97.6	95.6	91.0	95.3
✓	✗	98.2	97.8	98.0	92.2	96.6
✗	✓	97.8	99.8	98.2	94.4	97.6
✓	✓	99.4	99.8	98.6	96.2	98.5

Table 4: Ablation study of geometric supervision and self-guided world modeling on LIBERO. Geo. denotes geometric supervision, and WM. denotes self-guided world modeling. Success rates (%) are reported. **Bold** indicates the best result.

Number of Token	Spatial	Object	Goal	Long	Avg.
1	97.2	98.8	98.2	90.2	96.1
4	99.0	98.4	97.4	95.2	97.5
8	99.4	99.8	98.6	96.2	98.5
16	99.4	99.4	97.6	92.4	97.2

Table 5: Ablation study on the number of learnable dynamics tokens on LIBERO. Success rates (%) are reported. **Bold** indicates the best result.

from 96.1% to 98.5%. The largest improvement occurred on LIBERO-Long, where the success rate increased from 90.2% to 96.2%. However, increasing the token number to 16 reduced the average success rate to 97.2%. We therefore used eight dynamics tokens in the final model. This non-monotonic trend suggests that the gain is not simply proportional to the number of dynamics tokens. Instead, a moderate number of tokens appears sufficient to represent diverse interaction dynamics, whereas additional tokens provide no further benefit to action learning.

To further explore how geometric supervision influences the visual context used for latent dynamics modeling, we visualize the middle-layer attention from each dynamics token to the main-view image tokens from the same frame in Figure 4. With geometric supervision, the dynamics tokens attend more consistently to interaction-relevant regions, including the robot end effector, the manipulated receptacle, and their surrounding spatial context. Without geometric supervision, the attention is more frequently dominated by isolated visually salient objects, such as the package on the table, and shows weaker correspondence with the robot-object interaction. These patterns suggest that geometric supervision helps organize the policy-derived representation space around manipulation-relevant spatial relationships, providing a more structured visual context for action-conditioned self-guided world modeling. Complete attention maps and the null-action

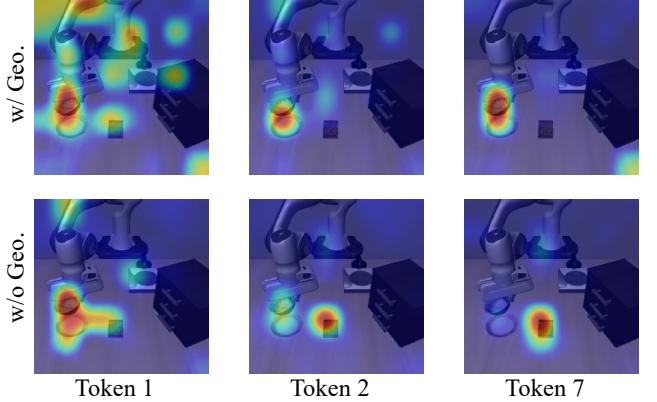


Figure 4: Representative middle-layer attention maps from dynamics tokens to main-view image tokens. The task is *Pick up the black bowl between the plate and the ramekin and place it on the plate*. With geometric supervision, the selected tokens attend more consistently to the robot-object interaction regions.

ablation are provided in Appendix.

5 Conclusion

This paper introduced SG-WAM, a self-guided world action model that learns action-conditioned future dynamics in a compact latent prediction space derived from the policy itself. SG-WAM introduces learnable dynamics tokens whose future states are predicted from the current policy state and intervening robot actions, while an EMA copy of the same policy provides stable future targets. A frozen 3D foundation model further shapes the policy’s main-view visual tokens with geometric structure, allowing latent future prediction and action generation to be jointly optimized over spatially grounded policy representations. The geometry teacher, SGWP, and EMA target pathway are used only during training and are removed at inference. Without large-scale embodied pretraining, the 0.9B-parameter SG-WAM achieves 98.5% average success on LIBERO and 73.0% on LIBERO-Plus, while outperforming representative explicit and latent WAM baselines in real-world evaluations. These results indicate that coupling self-guided future prediction with geometric grounding provides an effective approach to learning robust policies. Future work will investigate scaling this self-guided world-modeling framework to larger policy backbones and broader cross-embodiment datasets, with the goal of supporting diverse embodiments and manipulation tasks.

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